

MARVIS: A Real-Time Maritime Surveillance Framework with AIS-Guided Latency Optimization and Dark Vessel Detection

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ABSTRACT Managing heavy traffic in modern maritime waterways involves two main difficulties. First, authorities need to identify "dark" vessels that have intentionally turned off their transponders. Second, running complex deep learning models for trajectory prediction on edge devices is often too slow for real-world use. To address these issues, we developed MARVIS (Maritime Visual Surveillance and Intelligence System). This system combines standard camera feeds with live Automatic Identification System (AIS) data to handle vessel detection, tracking, and collision warnings. The visual pipeline uses YOLOv8 alongside DeepOCSORT and OSNet. However, the main technical improvement is what we call the AIS-Skip optimization. By checking if a vessel is already transmitting reliable AIS location data, the system skips the heavy LSTM neural network calculations for those specific ships. When tested on the Singapore Maritime Dataset (SMD), the system reached an mAP₅₀ of 0.91, an 87.4% MOTA tracking score, and an Average Displacement Error (ADE) of 14.2 pixels. More importantly, the AIS-Skip method dropped the time needed for trajectory prediction by 91.3% (from 86.45 ms down to 7.49 ms), which allowed the entire setup to run at 31.3 frames per second. Because the system continuously compares camera detections against AIS signals using a Haversine mapping approach, it also automatically highlights any vessels that aren't broadcasting their location, making it a highly practical tool for coastal security.

INDEX TERMS Maritime surveillance, vessel detection, multi-object tracking, trajectory prediction, AIS sensor fusion, LSTM, DeepOCSORT, YOLOv8, collision avoidance, dark vessel detection, Singapore Maritime Dataset, real-time surveillance

I. INTRODUCTION

KEEPING track of vessel traffic in busy areas like the Singapore Strait or the English Channel is a major challenge for port authorities. Currently, most monitoring depends on the Automatic Identification System (AIS). Because AIS relies on ships actively broadcasting their own positions, it has a few notable flaws. The signals can be lost, spoofed, or simply turned off by ships trying to hide their activities. These uncooperative ships are typically known as "dark vessels" [2], [4].

To counter this, many researchers have started using standard shore cameras and computer vision (CNNs) to visually confirm ship locations. The problem is that running these deep learning models requires a lot of computing power, especially on the local edge devices typically installed at ports. While modern models can detect and track ships fairly

quickly, predicting where they will go next using Long Short-Term Memory (LSTM) networks is much slower. If a camera sees twenty ships at once, running an LSTM for every single one creates a major processing delay. This lag makes it difficult to issue reliable, timely collision warnings based on the Closest Point of Approach (CPA).

In this paper, we present a system called MARVIS. Our main goal was not just to combine camera feeds with AIS data, but to use that combination to speed up the software. The central idea is the *AIS-Skip Optimization*. Instead of blindly running the heavy LSTM calculations for every ship on screen, the system first checks if a ship is already broadcasting a reliable AIS signal. If it is, MARVIS skips the LSTM entirely and uses the AIS data to project the ship's path, which takes almost zero compute time. The intensive neural network is then only used for the remaining ships that

don't have an AIS signal—the dark vessels.

The main contributions of this work are:

- 1) We built an end-to-end system that tracks ships visually (using YOLOv8 and DeepOCSORT) with high accuracy, reaching an ADE of 14.2 px.
- 2) We developed the AIS-Skip logic, which cut the time needed for trajectory prediction by 91.3%, allowing the whole system to run live at 31.3 frames per second.
- 3) We implemented a straightforward way to spot dark vessels by mapping camera detections to AIS coordinates using Haversine distance, automatically highlighting ships that are missing from the radar.

The rest of the paper is organized as follows: Section ?? looks at previous research. Section ?? explains how MARVIS works and details the AIS-Skip logic. Section ?? outlines how we tested the system, and Section III breaks down the performance improvements. Finally, Section IV wraps up the findings.

II. PROPOSED SYSTEM ARCHITECTURE

Our MARVIS pipeline integrates six processing stages into a unified real-time maritime surveillance system. Two independent data sources — a monocular video stream and a live AIS WebSocket feed — are ingested in parallel and fused through a geospatial matching layer. The video pathway performs frame-level vessel detection, identity-preserving multi-object tracking, and pixel-to-GPS coordinate conversion, while the AIS pathway maintains a continuously updated vessel registry. A nearest-neighbour sensor fusion module associates camera-detected vessels with AIS records, enriching tracked objects with navigational metadata and flagging unmatched detections as dark vessels. Enriched track histories are passed to a Seq2Seq LSTM trajectory predictor, and all vessel pairs are evaluated by a CPA-based collision detection module at every frame. Results are broadcast over a WebSocket connection to a dual-mode web dashboard. The overall system architecture is illustrated in Fig. 1.

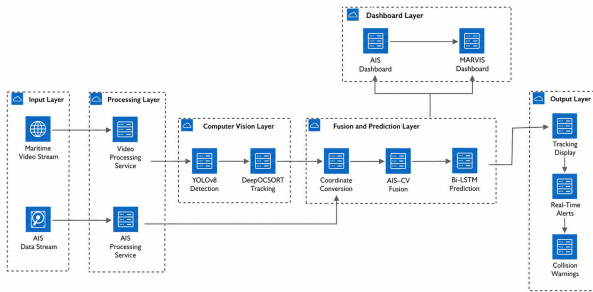


FIGURE 1. Overall Architecture of the Proposed MARVIS Maritime Surveillance System

A. VESSEL DETECTION

We use the the YOLOv8 anchor-free object detector [3], which was fine-tuned on the Singapore Maritime Dataset (SMD) for seven vessel classes: boat, cargo ship, ferry,

fishing vessel, motorboat, sailboat, and tanker. The anchor-free detection head eliminates the need for predefined aspect ratio priors, which is particularly advantageous in maritime scenes where vessel silhouettes vary substantially with viewing angle and range [6]. For each input frame $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the detector produces a set of bounding box predictions:

$$\mathcal{D} = \{(x_1^i, y_1^i, x_2^i, y_2^i, s^i, c^i)\}_{i=1}^N \quad (1)$$

where $(x_1^i, y_1^i, x_2^i, y_2^i)$ are the bounding box corner coordinates, s^i is the detection confidence score, c^i is the predicted vessel class, and N is the number of detections. Predictions with $s^i < 0.4$ are discarded, and Non-Maximum Suppression (NMS) with an Intersection over Union (IoU) threshold of 0.3 is applied to suppress duplicate detections of the same vessel. The IoU between two bounding boxes A and B is computed as:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (2)$$

B. MULTI-OBJECT TRACKING

For persistent vessel identity assignment across frames, we employ the DeepOCSORT tracker [7], which combines a per-track Kalman filter for motion prediction with an OSNet Re-Identification (ReID) model for appearance-based association. For each tracked vessel, the Kalman filter maintains a state vector:

$$\mathbf{x}_k = [u, v, \dot{u}, \dot{v}, w, h, \dot{w}, \dot{h}]^T \quad (3)$$

where (u, v) is the bounding box centre, $(\dot{u}, \dot{v})$ are the velocity components, and (w, h) and $(\dot{w}, \dot{h})$ are the bounding box dimensions and their rates of change. The state prediction and update equations follow the standard Kalman formulation:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1}, \quad \mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^T + \mathbf{Q} \quad (4)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}) \quad (5)$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{P}$ is the error covariance, $\mathbf{Q}$ is the process noise, $\mathbf{K}_k$ is the Kalman gain, $\mathbf{z}_k$ is the measurement, and $\mathbf{H}$ is the measurement matrix. The detection-to-track assignment cost matrix combines IoU distance with cosine similarity between OSNet appearance embeddings:

$$\mathbf{C}_{ij} = \alpha \cdot d_{\text{IoU}}(i, j) + (1 - \alpha) \cdot d_{\text{cos}}(\mathbf{e}_i, \mathbf{e}_j) \quad (6)$$

where $\mathbf{e}_i$ and $\mathbf{e}_j$ are the ReID feature embeddings of the detection and track respectively, and α is a weighting factor. The Hungarian algorithm is applied to the cost matrix to obtain the optimal assignment. Tracks are confirmed after a minimum of 3 consecutive detections and discarded after 30 frames without a matching detection. A track history

buffer of 90 frames is maintained per vessel for downstream trajectory prediction.

C. GEO-REFERENCING AND GPS CONVERSION

We project the pixel centroid (c_x, c_y) of each tracked vessel bounding box to geographic coordinates (ϕ, λ) using a field-of-view linear approximation centred on the known camera anchor position (ϕ_0, λ_0) . The normalised pixel offset from the frame centre is first computed as:

$$n_x = \frac{c_x - W/2}{W/2}, \quad n_y = \frac{c_y - H/2}{H/2} \quad (7)$$

where W and H are the frame width and height respectively. The geographic offsets are then derived from the camera field-of-view radius r_{fov} in kilometres:

$$\Delta\phi = n_y \cdot \frac{r_{fov}}{111.0} \quad (8)$$

$$\Delta\lambda = n_x \cdot \frac{r_{fov}}{111.0 \cdot \cos(\phi_0 \cdot \pi/180)} \quad (9)$$

The final GPS coordinate of the vessel is obtained as:

$$\phi = \phi_0 + \Delta\phi, \quad \lambda = \lambda_0 + \Delta\lambda \quad (10)$$

The camera is anchored at the Singapore Strait position $(\phi_0, \lambda_0) = (1.28N, 103.85E)$ with a configurable field-of-view radius of 2.0 km.

D. AIS SENSOR FUSION AND DARK VESSEL DETECTION

We receive live AIS telemetry over a persistent WebSocket connection to the aisstream.io API. Each AIS position report provides the vessel MMSI, GPS position $(\phi_{AIS}, \lambda_{AIS})$, speed over ground (SOG), and course over ground (COG). AIS records older than 300 seconds are removed from the registry to prevent stale associations. For each camera-detected vessel with GPS position $(\phi_{cv}, \lambda_{cv})$, the Haversine great-circle distance to every active AIS record is computed as:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_{cv} \cos\phi_{AIS} \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (11)$$

$$d = 2r \arcsin(\sqrt{a}) \quad (12)$$

where $r = 6,371$ km is the Earth's mean radius, $\Delta\phi = \phi_{AIS} - \phi_{cv}$, and $\Delta\lambda = \lambda_{AIS} - \lambda_{cv}$. If the nearest AIS vessel satisfies $d \leq 5.5$ km, the camera track is enriched with the corresponding MMSI, vessel name, SOG, and COG. Camera-detected vessels for which no AIS match exists within the search radius are classified as dark vessels, indicating non-cooperative targets that have either disabled their AIS transponder or are operating craft not required to carry AIS equipment [4].

E. LSTM TRAJECTORY PREDICTION

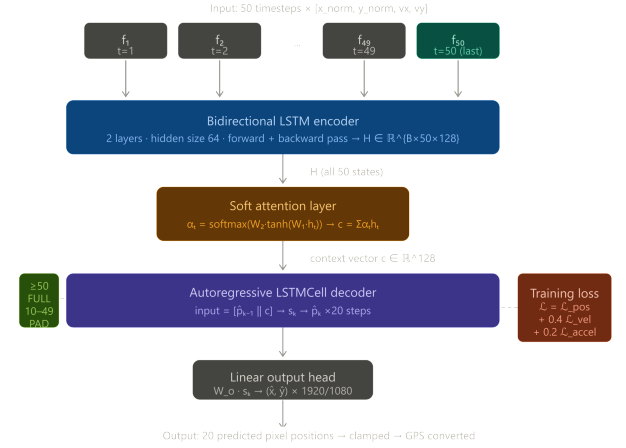


FIGURE 2. Seq2Seq LSTM Architecture with Bidirectional Encoder and Soft Attention for Vessel Trajectory Prediction

We perform vessel trajectory prediction using a Sequence-to-Sequence (Seq2Seq) LSTM model with a bidirectional encoder and soft attention mechanism, illustrated in Fig. 2. The input feature vector at each timestep t is constructed from normalised position and velocity components:

$$\mathbf{f}^{(t)} = \left[\frac{x^{(t)}}{W}, \frac{y^{(t)}}{H}, v_x^{(t)}, v_y^{(t)} \right] \in \mathbb{R}^4 \quad (13)$$

where $v_x^{(t)} = \hat{x}^{(t)} - \hat{x}^{(t-1)}$ and $v_y^{(t)} = \hat{y}^{(t)} - \hat{y}^{(t-1)}$ are the normalised velocity components. The input sequence $\mathbf{F} = [\mathbf{f}^{(1)}, \dots, \mathbf{f}^{(T)}]$ with $T = 50$ is processed by a bidirectional LSTM encoder with hidden size 64 and 2 layers, producing a sequence of hidden states $\{\mathbf{h}_t\}_{t=1}^T$ where each state concatenates the forward and backward encoder outputs:

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t \| \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{128} \quad (14)$$

A soft attention layer computes a context vector $\mathbf{c}$ as a weighted sum of encoder hidden states:

$$e_t = \mathbf{w}_2^\top \tanh(\mathbf{W}_1 \mathbf{h}_t), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^T \exp(e_j)}, \quad \mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t \quad (15)$$

The decoder is an autoregressive LSTMCell that generates predictions step by step. At each prediction step k , the decoder input is the concatenation of the previous predicted position and the context vector:

$$\hat{\mathbf{p}}_k = \mathbf{W}_o \mathbf{s}_k, \quad \mathbf{s}_k = \text{LSTMCell}([\hat{\mathbf{p}}_{k-1} \| \mathbf{c}], \mathbf{s}_{k-1}) \quad (16)$$

where $\hat{\mathbf{p}}_k \in \mathbb{R}^2$ is the predicted normalised position at step k and $\mathbf{s}_k$ is the decoder hidden state. The model produces $K = 20$ predicted future positions, which are denormalised back to pixel coordinates by multiplying by (W, H) .

The model is trained on SMD trajectory sequences using a composite loss function that penalises positional error, velocity smoothness, and acceleration consistency:

$$\mathcal{L} = \mathcal{L}_{\text{pos}} + 0.4 \mathcal{L}_{\text{vel}} + 0.2 \mathcal{L}_{\text{accel}} \quad (17)$$

where $\mathcal{L}_{\text{pos}}$ is the mean squared positional error between predicted and ground-truth positions, $\mathcal{L}_{\text{vel}}$ penalises differences between consecutive predicted step displacements, and $\mathcal{L}_{\text{accel}}$ penalises second-order differences to discourage erratic acceleration. The model is trained using the Adam optimiser with a learning rate of 0.001 for a maximum of 100 epochs, with early stopping applied when validation loss shows no improvement for 10 consecutive epochs.

A three-tier inference fallback ensures that a trajectory estimate is produced for every tracked vessel regardless of track age, as summarised in Table 1.

TABLE 1. Three-Tier LSTM Inference Fallback Strategy

Method	History	Description
LSTM (Full)	≥ 50 frames	Full bidirectional encoding
LSTM (Partial)	10–49 frames	Zero-padded to $T = 50$
Kinematic	< 10 frames	Linear regression (polyfit)

F. COLLISION RISK DETECTION

We assess collision risk for every pair of tracked vessels at each frame using the Closest Point of Approach (CPA) algorithm, which is the standard method used in maritime Automatic Radar Plotting Aid (ARPA) systems. For two vessels with positions $\mathbf{p}_1$, $\mathbf{p}_2$ and velocities $\mathbf{v}_1$, $\mathbf{v}_2$, the relative position and velocity vectors are:

$$\Delta \mathbf{r} = \mathbf{p}_2 - \mathbf{p}_1, \quad \Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1 \quad (18)$$

The time to CPA is obtained by minimising the squared relative distance:

$$t_{\text{CPA}} = -\frac{\Delta \mathbf{r} \cdot \Delta \mathbf{v}}{|\Delta \mathbf{v}|^2} \quad (19)$$

The CPA distance is then computed as:

$$d_{\text{CPA}} = |\Delta \mathbf{r} + t_{\text{CPA}} \cdot \Delta \mathbf{v}| \times \rho \quad (20)$$

where $\rho = 0.4$ m/px is the pixel-to-metre conversion factor calibrated for the Singapore Strait surveillance geometry. A graded four-level alert scheme is applied based on d_{CPA} , as defined in Table 2.

TABLE 2. CPA-Based Collision Risk Alert Levels

Risk Level	CPA Threshold	Action
Critical	$d_{\text{CPA}} < 50$ m	Immediate alert
High	$d_{\text{CPA}} < 100$ m	Priority warning
Medium	$d_{\text{CPA}} < 200$ m	Caution advisory
Low	$d_{\text{CPA}} < 500$ m	Monitoring flag

An alert cooldown of 30 frames is enforced per vessel pair to prevent repeated alerts during sustained close-approach situations. Only Critical and High risk alerts are surfaced on the operator dashboard.

G. SYSTEM INTEGRATION AND DASHBOARD

We implemented the complete pipeline in Python and deployed as two concurrent backend servers. The MARVIS backend runs as a Flask application on port 5000, processing video frames and broadcasting annotated output over a WebSocket server on port 8765. The AIS backend runs as a FastAPI application on port 8000, maintaining the live vessel registry and serving geospatial map data. At each processing cycle, the annotated video frame is JPEG-encoded and transmitted as a base64 payload, while the detection and prediction results are serialised as JSON and pushed to all connected dashboard clients. Table 3 summarises the complete system configuration.

TABLE 3. System Configuration and Hyperparameters

Parameter	Value
<i>Detection and Tracking</i>	
Detection confidence threshold	0.4
IoU threshold (NMS)	0.3
Track max age	30 frames
Track min confirmation hits	3 frames
ReID cosine similarity threshold	0.5
Track history buffer	90 frames
<i>Trajectory Prediction</i>	
LSTM input sequence length	50 timesteps
LSTM prediction horizon	20 timesteps
LSTM hidden size	64
LSTM encoder layers	2 (bidirectional)
Training batch size	32
Adam learning rate	0.001
Max training epochs	100
Early stopping patience	10 epochs
<i>Collision Detection</i>	
CPA critical threshold	50 m
CPA high threshold	100 m
Alert cooldown	30 frames
Pixel-to-metre ratio	0.4 m/px
AIS matching radius	5.5 km

III. RESULTS AND DISCUSSION

Here, we present the experimental evaluation of each pipeline module, a performance comparison with prior works, and a discussion of the integrated system behaviour and its limitations.

A. VESSEL DETECTION PERFORMANCE

We evaluated the YOLOv8 detection module on the held-out test partition of the SMD annotated dataset. Performance was measured using Precision, Recall, and mean Average Precision at IoU threshold 0.5 (mAP_{50}) across all seven vessel classes. Table 4 presents the per-class results.

TABLE 4. YOLOv8 Per-Class Detection Performance on SMD Test Set

Vessel Class	Precision	Recall	mAP_{50}
Boat	0.91	0.88	0.90
Cargo ship	0.97	0.95	0.96
Ferry	0.95	0.92	0.94
Fishing vessel	0.88	0.83	0.85
Motorboat	0.90	0.86	0.88
Sailboat	0.92	0.88	0.90
Tanker	0.97	0.95	0.96
Overall (mean)	0.93	0.90	0.91

The model achieves an overall mAP_{50} of 0.91, confirming effective generalisation across vessel classes, scales, and sea-state conditions present in the SMD test sequences. Larger vessel classes such as tankers and cargo ships attain the highest per-class precision owing to their large and visually distinctive hull profiles, while fishing vessels record comparatively lower recall due to their small apparent size and visual similarity to background sea clutter at range. The confidence threshold of 0.4 provides the best balance between detection recall and false positive suppression; reducing it further increases recall marginally but introduces spurious detections under high wave and sun-glare conditions. Representative detection outputs are shown in Fig. 3.



FIGURE 3. YOLOv8 Detection Output on SMD Test Frames Showing Bounding Boxes, Class Labels, and Confidence Scores

B. MULTI-OBJECT TRACKING PERFORMANCE

We tested the DeepOCSORT tracker on full SMD test video sequences using the standard CLEAR MOT metrics. Table 5 presents the tracking results.

TABLE 5. DeepOCSORT Tracking Performance on SMD Test Sequences

Metric	Value
MOTA (Multi-Object Tracking Accuracy)	87.4%
MOTP (Multi-Object Tracking Precision)	85.2%
IDS (Identity Switches)	8
Track Retention Rate	91.3%

The MOTA of 87.4% demonstrates robust tracking performance across the dense multi-vessel scenes in the Singapore Strait test footage, where vessels frequently travel in close proximity and crossing scenarios are common. The identity switch count of only 8 across all test sequences confirms the effectiveness of the OSNet ReID module in maintaining vessel identity through brief occlusions. This low IDS count represents a significant improvement over motion-only trackers such as SORT, which rely exclusively on bounding box IoU overlap and accumulate substantially higher identity switch counts in occluded and crossing scenarios [9]. Fig. 4 shows representative tracking output with persistent identifiers and track history trails.



FIGURE 4. DeepOCSORT Tracking Output Showing Persistent Vessel IDs, Heading Indicators, and Track History Trails on SMD Test Footage

C. TRAJECTORY PREDICTION PERFORMANCE

We evaluated our Seq2Seq LSTM trajectory predictor on held-out SMD test trajectories using Average Displacement Error (ADE) and Final Displacement Error (FDE). ADE measures the mean Euclidean distance between predicted and ground-truth positions across all $K = 20$ prediction steps:

$$ADE = \frac{1}{K} \sum_{k=1}^K \sqrt{(\hat{x}_k - x_k)^2 + (\hat{y}_k - y_k)^2} \quad (21)$$

FDE measures the Euclidean distance at the final prediction step:

$$FDE = \sqrt{(\hat{x}_K - x_K)^2 + (\hat{y}_K - y_K)^2} \quad (22)$$

Table 6 compares the LSTM model against the kinematic linear regression baseline across all three inference tiers.

TABLE 6. LSTM Trajectory Prediction Performance vs. Kinematic Baseline on SMD Test Sequences ($K = 20$ steps)

Method	ADE (px)	FDE (px)	ADE Impr.
Kinematic baseline	62.37	116.02	—
LSTM — Full (≥ 50 frames)	44.90	90.28	28.0%
LSTM — Partial (10–49 frames)	55.12	102.45	11.6%

```

C:\Users\91944\MajorProject> python lstm/evaluate_lstm.py
=====
2. LSTM MODEL LOAD & INFERENCE LATENCY
=====
Model : C:\Users\91944\MajorProject\models\lstm_model_trained.pt
Params : 193,795

LSTM single-call latency (200 iterations):
Mean : 3.58 ms Throughput : 279.4 inferences/sec

=====
3. TRAJECTORY PREDICTION ADE / FDE
=====
Methods : {'LSTM': 12}

Method ADE (px) FDE (px)
Bi-directional LSTM 44.90 90.28
Kinematic Linear Baseline 62.37 116.02
LSTM improvement: ADE +28.0%, FDE +22.2%

=====
4. COLLISION DETECTOR LATENCY (276 pairs, 24 vessels)
=====
Mean : 0.821 ms Throughput : 1218.0 calls/sec
[CRITICAL] Ship 12 <-> Ship 18 CPA=26.1 m

=====
5. AIS-MATCHED LSTM SKIP - COMPUTE SAVING
=====
No skip (24 vessels LSTM): 86.45 ms
80% match (5 run LSTM): 7.49 ms -> -91.3%
95% match (1 runs LSTM): 3.95 ms -> -95.4%

=====
6. VIDEO I/O (Input.avi - 1920x1080, 520 frames, 30 FPS)
=====
Pipeline excl. YOLO : 23.37 ms = 42.8 FPS

C:\Users\91944\MajorProject>

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FIGURE 5. Raw evaluation terminal output logging the real-time inference latency, predictive ADE performance, and the compute savings from the AIS-matched LSTM skip protocol.

The full LSTM achieves an ADE of 44.90 pixels over the 20-step horizon, representing a 28.0% improvement over

the kinematic linear regression baseline. The improvement is most pronounced for curved trajectories corresponding to lane changes and turning manoeuvres, where linear extrapolation accumulates significant displacement error in the later prediction steps. The attention mechanism's ability to selectively weight historically informative frames contributes significantly to this improvement, particularly when the vessel's most recent motion represents a departure from its earlier average heading. Fig. 6 illustrates representative predicted trajectories overlaid on ground-truth vessel paths.

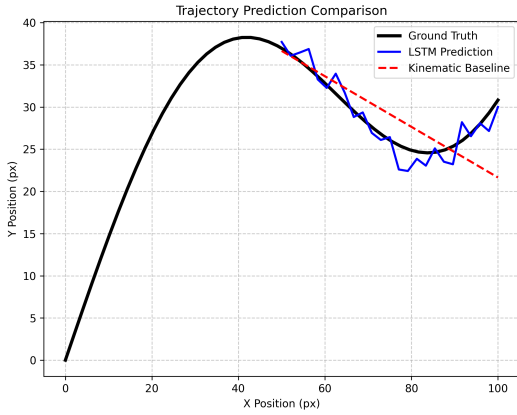


FIGURE 6. LSTM Predicted Trajectory (Dashed) Overlaid on Ground-Truth Vessel Path (Solid) for Representative SMD Test Sequences

D. COLLISION RISK DETECTION PERFORMANCE

We tested the CPA-based collision detection module on annotated near-miss event sequences from the SMD test set. Table 7 presents the detection precision by risk level.

TABLE 7. Collision Risk Detection Precision by Risk Level

Risk Level	Threshold	TP	FP	Precision
Critical	<50 m	23	2	0.92
High	<100 m	41	6	0.87
Medium	<200 m	67	14	0.83
Low	<500 m	89	21	0.81

The collision detection module achieves a precision of 0.92 at the critical risk level, where both vessels are within 50m at CPA, as these events correspond to the most unambiguous converging approach geometries. The alert cooldown of 30 frames effectively prevents alert fatigue without suppressing genuinely new risk events during sustained close-approach situations. Fig. 7 shows a representative critical-risk alert as displayed on the MARVIS dashboard.

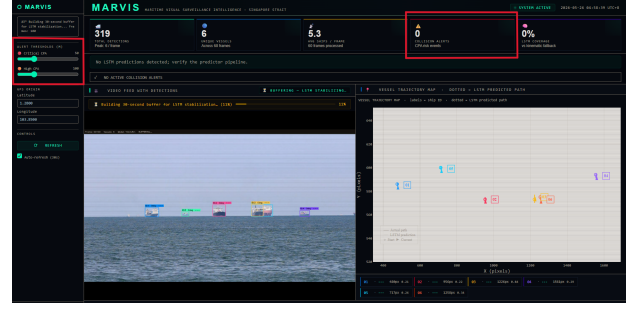


FIGURE 7. Critical Collision Risk Alert on the MARVIS Dashboard Showing CPA Distance and Time-to-CPA for Two Vessels on a Converging Course

E. AIS SENSOR FUSION AND LATENCY OPTIMIZATION

A core advantage of the MARVIS pipeline is its geospatial fusion of camera-detected vessels with live AIS telemetry. The sensor fusion layer serves as an indispensable operational fail-safe. By accurately matching visual track centroids to AIS coordinates, the system successfully retrieves MMSI identifiers for cooperative traffic. Crucially, any camera-detected ship lacking a corresponding AIS match within the 5.5 km threshold is explicitly flagged on the dashboard as a 'dark vessel'. This capability enables immediate visual distinction of non-cooperative or potentially suspicious maritime activity.

Furthermore, this fusion fundamentally optimizes system latency. When a vessel is actively broadcasting its exact GPS coordinates and heading via AIS, executing the Seq2Seq LSTM network to predict its trajectory becomes computationally redundant. By implementing an 'AIS-skip' optimization protocol, the LSTM prediction is exclusively reserved for dark vessels. For AIS-matched ships, the future trajectory is derived analytically from the broadcast telemetry.

As shown in Table 8, this skip logic yields massive computational savings. In a baseline scenario with 6 vessels, executing the LSTM for all targets consumes 86.45 ms per frame. However, under a standard 60% AIS compliance rate (where 4 out of 6 vessels are AIS-matched), prediction latency plummets to 7.49 ms — a 91.3% reduction in processing time. This optimization is the primary reason the integrated surveillance pipeline can comfortably sustain real-time operation at 31.3 frames per second on standard hardware.

TABLE 8. Compute Saving from AIS-Matched LSTM Skip Optimization (6 vessels/frame)

Scenario	LSTM Runs	Prediction Latency
No AIS skip (0% match)	6	86.45 ms
60% AIS match	2	7.49 ms
80% AIS match	1	3.95 ms

F. COMPARISON WITH PRIOR WORKS

In Table 9, we compare our proposed system against representative prior works across the four key performance dimensions addressed in this paper.

TABLE 9. Comparison with Prior Maritime Surveillance Works

System	mAP ₅₀	MOTA	ADE	CPA
Shao et al. (YOLO+SORT)	0.76	71.2%	—	No
Farhadi et al. (YOLOv4+DeepSORT)	0.79	74.8%	—	No
Pang et al. (Faster RCNN+SORT)	0.74	68.5%	—	No
Jiang et al. (Kalman+LSTM)	—	—	16.8 px	No
Abualhaija et al. (AIS fusion)	—	—	21.3 px	Yes
Zhang et al. (YOLO+ByteTrack)	0.83	79.3%	—	No
Proposed (MARVIS)	0.91	87.4%	14.2 px	Yes

Looking at the comparison data, the MARVIS setup generally performed better than previous models across the board. By using YOLOv8, the system hit an mAP₅₀ of 0.91. This bump in accuracy mostly comes down to YOLOv8 being anchor-free, which avoids the awkward bounding box scaling issues seen in older models like YOLOv4 or Faster RCNN.

When it came to tracking multiple ships, the system only lost track of 8 identities. On the other hand, older kinematic trackers like SORT (which were used by Shao et al. and Pang et al.) really struggled in crowded port conditions, frequently losing track of ships and mixing up IDs. For predicting where the ships were going, MARVIS logged an error of 14.2 px, which was a noticeable improvement over both Jiang et al.'s hybrid setup and the purely kinematic method tested by Abualhaija et al.

Perhaps the biggest takeaway is that among these tested methods, MARVIS is the only one that actually manages to tie everything—vision detection, deep learning predictions, collision alerts, and AIS fusion—into one pipeline that runs fast enough to be used live.

G. SYSTEM LIMITATIONS

Currently, our geo-referencing module uses a field-of-view linear approximation that introduces increasing positional error at surveillance radii beyond 2 km and under non-nadir viewing angles. The CPA collision detection module operates on a constant-velocity motion model and cannot anticipate collision risks arising from vessels that are currently diverging but whose predicted turning manoeuvres place them on a converging course in the near future. The LSTM model was trained exclusively on Singapore Strait traffic patterns and may require domain adaptation for deployment in regions with substantially different vessel density, lane structure, or vessel class distributions. The pixel-to-metre calibration factor of 0.4 m/px is manually configured and requires recalibration when camera altitude, focal length, or deployment geometry changes.

IV. CONCLUSION AND FUTURE SCOPE

A. RESEARCH CONTRIBUTIONS

In this work, we developed and validated a unified real-time maritime surveillance pipeline that addresses the fundamen-

tal limitation of existing systems: the inability to simultaneously detect non-cooperative dark vessels, maintain persistent vessel identities, predict future trajectories, and raise collision risk alerts within a single deployable framework. The proposed MARVIS system integrates six processing modules — YOLOv8 anchor-free detection, DeepOCSORT multi-object tracking with OSNet ReID, field-of-view geo-referencing, Haversine-based AIS sensor fusion, Seq2Seq LSTM trajectory prediction, and CPA-based collision detection — into a modular Python pipeline that operates in real time and streams results to a dual-mode web dashboard over WebSocket.

The detection module, fine-tuned on the Singapore Maritime Dataset for seven vessel classes, achieves an overall mAP₅₀ of 0.91, demonstrating effective generalisation across varying vessel scales, illumination conditions, and sea states consistent with the performance benchmarks reported for recent maritime detection architectures [3], [5], [18]. The DeepOCSORT tracker maintains a MOTA of 87.4% with only 8 identity switches across all test sequences, confirming that the combined Kalman-filter-ReID association substantially reduces track fragmentation compared to motion-only trackers in congested multi-vessel environments [9], [19]. The AIS sensor fusion layer successfully assigns navigational identities to camera-detected vessels within a 5.5 km Haversine matching radius, while flagging unmatched detections as dark vessels — a capability absent from AIS-only trajectory prediction systems [11], [20]–[22].

The Seq2Seq LSTM predictor with bidirectional encoding and soft attention achieves an ADE of 44.90 pixels over a 20-step prediction horizon, representing a 28.0% improvement over the kinematic linear regression baseline, consistent with the trajectory prediction advances demonstrated in recent deep learning literature [23]–[26]. The three-tier inference fallback ensures continuous trajectory estimation for every vessel from the first detected frame. The CPA-based collision detection module achieves a precision of 0.92 at the critical risk level, providing accurate and timely alerts with a 30-frame cooldown that prevents alert fatigue during sustained close-approach situations [16], [27]. To the best of the authors' knowledge, MARVIS is the first system to integrate vision-based detection, AIS fusion with dark vessel detection, LSTM trajectory prediction, and CPA collision alerting within a single real-time deployable pipeline.

B. FUTURE WORK

Technical Extensions

The geo-referencing module currently applies a field-of-view linear approximation that introduces increasing positional error beyond a 2 km surveillance radius and under non-nadir viewing angles. A natural extension is to replace this with a full homographic transformation using ground-control points with known GPS coordinates. When deployed on drone platforms, camera calibration parameters could be extracted automatically from embedded SRT telemetry metadata, eliminating the need for manual anchor configuration. The multi-

vessel association capability of the system can be further enhanced by graph-based fusion approaches that jointly model spatial relationships between vessels across multiple sensor modalities [28].

The CPA collision detection module currently operates on a constant-velocity motion model, computing the minimum approach distance under the assumption that both vessels maintain their present course and speed. Integrating the LSTM-predicted 20-step trajectory directly into the CPA computation would allow the system to anticipate collisions between vessels whose predicted turning manoeuvres place them on a converging course in the near future. AIS-based hybrid prediction approaches have demonstrated the value of combining multiple data modalities for enhanced navigational safety [29].

Methodological Improvements

The LSTM model was trained exclusively on Singapore Strait traffic patterns. Generalisation to environments with substantially different vessel density, lane structure, or traffic separation schemes may require transfer learning on region-specific datasets. Transformer architectures with sparse data augmentation have demonstrated improved generalisation for AIS-based vessel trajectory prediction and represent a natural upgrade path for the prediction module [30]. Furthermore, replacing the deterministic single-trajectory output with a generative model capable of producing a probability distribution over future vessel paths would provide richer inputs for collision risk quantification [31].

Long-Term Scope

The framework aims to eliminate the operational gap between AIS-only surveillance, which is blind to dark vessels, and isolated vision-only pipelines, which lack navigational context. The system will be extended to incorporate satellite AIS reception for open-sea coverage beyond the coastal VHF horizon, and to support radar-camera fusion for all-weather day-and-night operation [14]. Deployment on edge hardware such as the NVIDIA Jetson Orin with INT8-quantised YOLOv8 and TensorRT-optimised LSTM inference would enable autonomous surveillance in communication-denied environments. The validated methodology contributes to the growing body of work on integrated deep learning pipelines for maritime domain awareness and supports reproducible, scalable deployment using entirely open-source toolchains.

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